

Receiver Design Project

Superheterodyne Receiver

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1 TUNING SOLUTION

PARAMETER	VALUE (MHz)	VALUE (α)	ATTENUATION
RF Band	1250 MHz - 1550 MHz		
f_0^{IF} (reasoning below)	320 MHz		
LO Band (see calc.)	1570 MHz - 1870 MHz		
Image Band (see calc.)	1890 MHz - 2190 MHz	1.883 - 3.351	40 dB Minimum
ω_{3a}^m (see calc.)	106.667 MHz	59.19	>60 dB
ω_{3b}^m (see calc.)	945.0 MHz - 1095.0 MHz	1.248 - 2.684	30 dB Minimum
ω_{3c}^m (see calc.)	625.0 MHz - 775.0 MHz	4.75 - 7.25	>60 dB
ω_{3d}^m (see calc.)	3460 MHz - 4060 MHz	8.667 - 10.94	>60 dB
ω_{3e}^m (see calc.)	2820 MHz - 3420 MHz	6.11 - 8.512	>60 dB
ω_{2a}^m (see calc.)	160.0 MHz	38.83	>60 dB
ω_{1a}^m (see calc.)	320 MHz	18.12	>60 dB

1.1 Selection of IF Center

We compared both high-side and low-side tuning solutions, and chose to go with high-side. Our method for doing this involved first plotting spurious sources and images for all potential IF center frequencies as their center frequency, shown in figure 1. This enabled us to select an IF center which balances the worst-case signals, the image $LO + IF$, and the ω_{3b} murphy signal $\frac{LO+IF}{2}$.

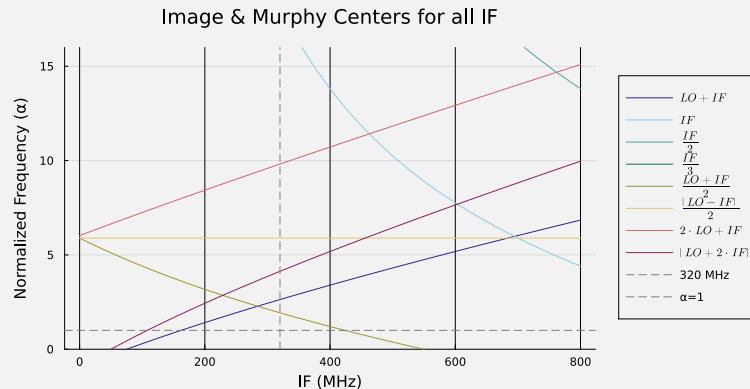


Figure 1: For any given IF, the lowest alpha for each signal is desired. 1α maps to 30 dB of attenuation, the minimum allowable attenuation. We select 320 MHz, as it offers improved image attenuation at the cost of attenuation in murphy signals.

2 IF Filter Specification

PARAMETER	VALUE
Order	3
Type	Butterworth
Kind	Band
Center	320 MHz
Bandwidth	300 MHz
In/Out Impedance	Matched to 50 Ω
Insertion Loss	1 dB
Max Input Power	10 W

2.1 Selection of IF Filter Order

Due to the narrow bandwidth, a butterworth filter is selected for its nearly linear phase-delay, and good dispersion. The FCC specification assigns our signals a 2.0 MHz bandwidth, which are no closer than 5.0 MHz. To determine a filter specification, we will express the spacing between stations in terms of its α , to select a minimum roll-off.

Let the upper and lower frequency bounds be denoted in terms of f_c ,

$$f_H(f_c) = f_c + 1 \text{ MHz} \quad \text{and} \quad f_L(f_c) = f_c - 1 \text{ MHz}$$

And the center frequency as

$$f_0(f_c) = \sqrt{f_H(f_c) \cdot f_L(f_c)}$$

Then we define:

$$\Delta(f_c) = \frac{f_0(f_c)}{f_H(f_c) - f_L(f_c)} \quad \alpha(f_c) = \left| \Delta(f_c) \cdot \left(\frac{f_c + \text{spacing}}{f_0(f_c)} - \frac{f_0(f_c)}{f_c + \text{spacing}} \right) \right| - 1 \quad (2.1)$$

We find that, at worst, we filter signals at 3.96α or further by 40 dB, which demands a 3rd order butterworth filter.

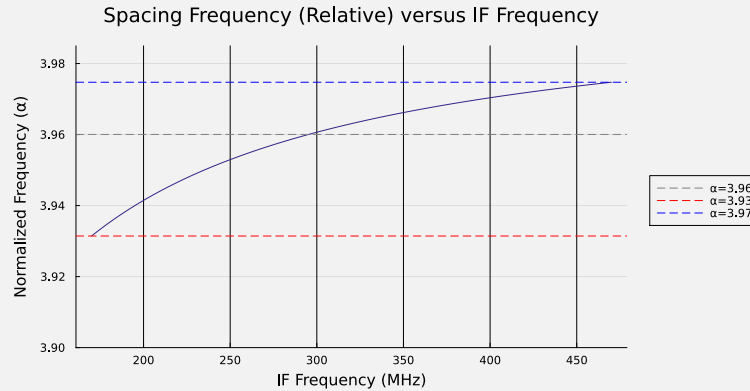


Figure 2: Filter response at $\alpha = f_c + 5 \text{ MHz}$.

3 Local Oscillator Specification

Parameter	Value
Carrier Frequency	1 GHz - 2 GHz
Carrier Power	13 dBm min.
Accuracy	At least ± 64 ppm between -25°C to 85°C
Harmonics and Spurs	-60 dBc maximum
Phase Noise	At most -110 dBc in a 1 Hz bandwidth at 1 kHz from the carrier frequency f_0
Noise Floor	-155 dBm in a 1 Hz bandwidth
DC Power	+7 volts to +24 volts at 2 W typical to 2.5 W maximum
Efficiency ^(see calc.)	0.01 typical (0.08 at max power)
Frequency Pulling (1.5:1 VSWR)	1 MHz peak-to-peak
Frequency Pushing	10 kHz/V max.

3.1 Selection of LO Accuracy

IF port accuracy is specified in ppm, as a function of the LO oscillator frequency and given in eq. 3.1, and can range from 1570 MHz - 1870 MHz, specified in section 1. LO frequency error may be no more than 10% of the IF bandwidth, as a rule of thumb.

The worst case scenario occurs at $f_{\text{LO}} = 1570$ MHz, with a value of 64 ppm ppm^(see calc.). Thus, our specification must have accuracy better than ± 64 ppm to ensure the ± 1 MHz frequency tolerance is met across the entire LO frequency range. Complete accuracy characteristics are shown in figure 3.

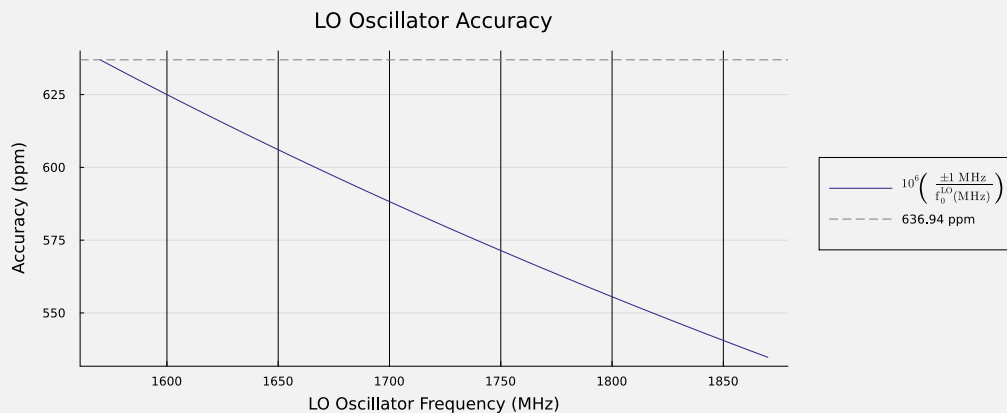


Figure 3: Complete LO oscillator accuracy characteristics. The worst case is given at the highest LO frequency, labelled by a dashed line.

4 Appendix: Calculations

4.1 LO Bandwidth

$$f_{LO,L} = f_{RF,L} + f_{IF} = 1250 + 320 = 1570 \text{ MHz}$$

$$f_{LO,H} = f_{RF,H} + f_{IF} = 1550 + 320 = 1870 \text{ MHz}$$

4.2 Normalized Frequency Parameter

The normalized frequency parameter α is defined as:

$$\% \Delta = \frac{f_0}{f_H - f_L} \quad (1)$$

$$\alpha(f) = \left| \% \Delta \left(\frac{f}{f_0} - \frac{f_0}{f} \right) \right| - 1 \quad (2)$$

4.3 Image Band

$$f_{image,L} = f_{LO,L} + f_{IF} = 1890 \text{ MHz} \quad (3)$$

$$f_{image,H} = f_{LO,H} + f_{IF} = 2190 \text{ MHz} \quad (4)$$

Range: $\alpha \in [1.883, 3.351]$

4.4 Murphy Spectrum - ω_{1a}^m

$$\omega_{1a}^m = f_{IF} = 320 \text{ MHz} \quad (5)$$

Range: $\alpha = 18.12$

4.5 Murphy Spectrum - ω_{2a}^m

$$\omega_{2a}^m = \frac{f_{IF}}{2} = 160 \text{ MHz} \quad (6)$$

Range: $\alpha = 38.83$

4.6 Third-Order Murphy Frequencies

4.6.1 Murphy Spectrum - ω_{3a}^m

$$\omega_{3a}^m = \frac{f_{IF}}{3} = 106.667 \text{ MHz} \quad (7)$$

Range: $\alpha = 59.19$

4.6.2 Murphy Spectrum - ω_{3b}^m

$$\omega_{3b,L}^m = \frac{f_{LO,L} + f_{IF}}{2} = 945 \text{ MHz} \quad (8)$$

$$\omega_{3b,H}^m = \frac{f_{LO,H} + f_{IF}}{2} = 1095 \text{ MHz} \quad (9)$$

Range: $\alpha \in [1.248, 2.684]$

4.6.3 Murphy Spectrum - ω_{3c}^m

$$\omega_{3c,L}^m = \frac{|f_{LO,L} - f_{IF}|}{2} = 625 \text{ MHz} \quad (10)$$

$$\omega_{3c,H}^m = \frac{|f_{LO,H} - f_{IF}|}{2} = 775 \text{ MHz} \quad (11)$$

Range: $\alpha \in [4.75, 7.25]$

4.6.4 Murphy Spectrum - ω_{3d}^m

$$\omega_{3d,L}^m = 2f_{LO,L} + f_{IF} = 3460 \text{ MHz} \quad (12)$$

$$\omega_{3d,H}^m = 2f_{LO,H} + f_{IF} = 4060 \text{ MHz} \quad (13)$$

Range: $\alpha \in [8.667, 10.94]$

4.6.5 Murphy Spectrum - ω_{3e}^m

$$\omega_{3e,L}^m = |2f_{LO,L} - f_{IF}| = 2820 \text{ MHz} \quad (14)$$

$$\omega_{3e,H}^m = |2f_{LO,H} - f_{IF}| = 3420 \text{ MHz} \quad (15)$$

Range: $\alpha \in [6.11, 8.512]$

4.6.6 LO Oscillator Accuracy (Worst Case)

$$\text{ppm}(\Delta f_0) \equiv 10^6 \left(\frac{0.1 \cdot \Delta f_r(\text{Hz})}{f_0(\text{Hz})} \right) = 10^6 \left(\frac{\pm 1 \text{ MHz}}{1570 \text{ MHz}} \right) \approx 64 \text{ ppm} \quad (16)$$

4.6.7 LO Oscillator Efficiency

$$\frac{10^{(13/10)} \text{ mW}}{2 \text{ W}} = \frac{20 \text{ mW}}{2000 \text{ mW}} = 0.01 \quad (17)$$

5 Appendix: Other

a.1

We also consider the low-side tuning solution for the receiver, which nets similar, but slightly worse results. As expected, the image and $2f_{LO} + f_{IF}$ are the most concerning RF signals. The best we could achieve for the image while attenuating this murphy signal by 30dB is approximately 1.65α from the RF signal, which is $\sim 13\%$ closer.

